

Executive Project Report: Real Estate Predictive Valuation Pipeline

1. Project Objectives

The objective of this analysis was to transition away from speculative property valuation by building a data-driven regression framework. Using a 545-row housing profile dataset, the model isolates structural drivers and outputs predictive valuations based on property characteristics.

2. Performance Metrics & Comparative Modeling

The dataset was split using an 80/20 train-test ratio. Two distinct algorithms were evaluated to find the best balance between predictive stability and variance capturing:

Machine Learning Model	Mean Absolute Error (MAE)	Root Mean Squared Error (RMSE)	R2 Score
Linear Regression (Winner)	970,043.40	1,324,506.96	0.6529
Random Forest Regressor	1,021,546.04	1,400,565.97	0.6119

Analysis: Due to the compact size of the dataset, the **Linear Regression** framework generalized much more effectively. The Random Forest variant overfit minor feature fluctuations, resulting in a higher error boundary (MAE of 1,021,546.04 vs 970,043.40) and a lower coefficient of determination (R^2).

3. Core Visual Discoveries

- **Distribution Skew:** The property valuation landscape is heavily right-skewed. The vast majority of housing is packed between 3,000,000 and 5,500,000, with a long, trailing tail representing high-end luxury estates upwards of 12,000,000.
- **Feature Interplay:** Total living area and modern plumbing systems (bathrooms) heavily dictate pricing tiers. Sheer bedroom count holds a back seat, highlighting a consumer preference for premium, wide layouts over crammed room structures.